INCH-POUND

A-A-59371
09 September 2003

COMMERCIAL ITEM DESCRIPTION
VOLTMETER

1 SCOPE. This Commercial Item Description (CID) describes a Direct Current (DC) Voltmeter. This CID is meant as a minimum requirement for the DC Voltmeter, in which only those manufacturers' products meeting or surpassing the following requirements are to be supplied per this CID.

2 SALIENT CHARACTERISTICS.

2.1 Equipment covered by this CID shall be commercially available equipment and may modified to the extent necessary to meet the following description. The equipment shall be Class 3, in accordance with MIL-PRF-28800. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

2.2 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.3 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 2000 hours of operation with a statistical certainly of 95%.

2.4 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611

AMSC N/A FSC 6625

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2.5 Electrical Power Sources and Connections. The equipment shall operate from nominal commercial, military, and shipboard power source of 120Vrms or 240Vrms single-phase AC (± 10 %) at line frequencies of 50 Hz and 60 Hz (± 5 %) as specified in MIL-PRF-28800.

2.6 Performance Characteristics.

2.6.1 Voltage Range. The equipment shall have the DC voltage range from 0 to 750 Volts.

2.6.2 Voltage Accuracy. The voltage accuracies and ranges shall be within those specified as below.

Ranges (Volts DC)	Accuracy (%)
0 to 150	0.75
0 to 300	0.75
0 to 750	0.75

2.6.3 Scale Division Quantity. The equipment shall have 100 single indicators all ranges.

2.6.4 Sensitive Rating. The equipment shall have a sensitivity of 1000 ohms per volt.

2.7 Movement Type. The movement type shall be permanent magnet moving-coil operation.

2.8 Circuit Attachment Method. Circuit attachment Method shall be 2 binding post.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE.

4.1 Product Conformance. The Products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerance using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PACKING. Preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES.

6.1 Sources of Documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 The Code of Federal Regulation (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

MILITARY INTEREST

Custodian:
Air Force – 99

Preparing activity:
Air Force - 84

Agent:
Air Force - 99

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